

A Countable-Generation Strengthening of the Laustsen–White Wark-Dual Example

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Source question and theorem

The source is Niels Jakob Laustsen and Jared T. White, *Subspaces that can and cannot be the kernel of a bounded operator on a Banach space*, arXiv:1811.02399v3. The paper asks which closed subspaces F of a Banach space E are kernels of bounded operators $E \rightarrow E$. It also uses this to produce a dual Banach algebra $\mathcal{B}(E)$ which is not weak-star topologically left Noetherian.

SUBSPACES THAT CAN AND CANNOT BE THE KERNEL OF A BOUNDED OPERATOR ON A BANACH SPACE

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ABSTRACT. Given a Banach space E , we ask which closed subspaces may be realised as the kernel of a bounded operator $E \rightarrow E$. We prove some positive results which imply in particular that when E is separable every closed subspace is a kernel. Moreover, we show that there exists a Banach space E which contains a closed subspace that cannot be realised as the kernel of any bounded operator on E . This implies that the Banach algebra $\mathcal{B}(E)$ of bounded operators on E fails to be weak*-topologically left Noetherian in the sense of [7]. The Banach space E that we use is the dual of one of Wark’s non-separable, reflexive Banach spaces with few operators.

1. INTRODUCTION

In this note we address the following natural question: given a Banach space E , which of its closed linear subspaces F are the kernel of some bounded linear operator $E \rightarrow E$? We shall begin by showing that if either E/F is separable, or F is separable and E has the separable complementation property, then F is indeed the kernel of some bounded operator

More precisely, let X be a non-separable reflexive Banach space with few operators in the sense that

$$\mathcal{B}(X) = \mathbb{C} \operatorname{id}_X + \mathcal{X}(X),$$

where $\mathcal{X}(X)$ is the ideal of operators with separable range. Wark constructed such spaces, and the source paper takes $E = X^*$. Laustsen–White prove that $\mathcal{B}(E)$ has a weak-star closed left ideal which is not weak-star topologically finitely generated.

Idea of the proof

The source proof uses a finite-intersection dichotomy: if a subspace $D \subseteq E = X^*$ is a finite intersection of kernels of operators on E , then either D or E/D is separable. The same proof survives for countably many kernels. The only change is that a countable closed linear span of separable ranges is still separable.

Laustsen–White choose a closed subspace $D \subseteq E$ such that both D and E/D are non-separable. The countable dichotomy therefore says that D is not a countable intersection of kernels. But if their weak-star closed left ideal

$$\mathcal{I}_D = \{T \in \mathcal{B}(E) : D \subseteq \ker T\}$$

were weak-star countably generated by $T_1, T_2, \dots$, then the intersection of the kernels of all operators in $\mathcal{I}_D$ would equal $\bigcap_n \ker T_n$. The left-hand side is exactly D , contradicting the countable dichotomy.

Definitions

Call a weak-star closed left ideal $\mathcal{I}$ of a dual Banach algebra A *weak-star topologically countably generated* if there are $a_1, a_2, \dots \in \mathcal{I}$ such that

$$\mathcal{I} = \overline{\text{span}\{ba_n : b \in A, n \in \mathbb{N}\}}^{w*}.$$

Lemma 1 (Countable kernel-intersection dichotomy). *Let X be a non-separable reflexive Banach space such that*

$$\mathcal{B}(X) = \mathbb{C} \text{id}_X + \mathcal{X}(X),$$

where $\mathcal{X}(X)$ consists of the operators with separable range. Put $E = X^$. If $T_1, T_2, \dots \in \mathcal{B}(E)$ and*

$$D = \bigcap_{n=1}^{\infty} \ker T_n,$$

then either D or E/D is separable.

Proof. Since X is reflexive, every $T_n \in \mathcal{B}(E) = \mathcal{B}(X^*)$ is the adjoint of some $S_n \in \mathcal{B}(X)$. Let

$$F = D_{\perp} = \{x \in X : f(x) = 0 \text{ for every } f \in D\}.$$

Using the standard annihilator identity,

$$F = \left(\bigcap_{n=1}^{\infty} \ker S_n^* \right)_{\perp} = \overline{\text{span}} \bigcup_{n=1}^{\infty} \text{im } S_n.$$

Write $S_n = \alpha_n \text{id}_X + R_n$, where $\alpha_n \in \mathbb{C}$ and $R_n \in \mathcal{X}(X)$.

If $\alpha_n = 0$ for every n , then F is the closed span of countably many separable subspaces, hence is separable. Since F is a closed subspace of the reflexive space X , it is reflexive, and therefore F^* is separable. The quotient E/D is naturally isomorphic to F^* , so E/D is separable.

If $\alpha_m \neq 0$ for some m , then for every $x \in X$,

$$x = \alpha_m^{-1}(S_mx - R_mx) \in F + \overline{\text{im } R_m}.$$

Thus X/F is a quotient of the separable space $\overline{\text{im } R_m}$, and hence X/F is separable. Since X/F is reflexive, its dual is separable. But D is naturally isomorphic to $(X/F)^*$, so D is separable. $\square$

Theorem 1. *Let X be one of Wark’s non-separable reflexive Banach spaces with few operators, and put $E = X^*$. Then $\mathcal{B}(E)$, with its canonical dual Banach algebra weak-star topology, has a weak-star closed left ideal which is not weak-star topologically countably generated.*

Proof. The source paper proves, using a Markushevich-basis argument, that E contains a closed subspace D such that both D and E/D are non-separable. Define

$$\mathcal{J}_D = \{T \in \mathcal{B}(E) : D \subseteq \ker T\}.$$

This is a left ideal. It is weak-star closed because, under the standard predual $E \widehat{\otimes} E^*$ of $\mathcal{B}(E)$,

$$\mathcal{J}_D = \{d \otimes \lambda : d \in D, \lambda \in E^*\}^\perp.$$

Assume, toward a contradiction, that $\mathcal{J}_D$ is weak-star topologically countably generated by $T_1, T_2, \dots \in \mathcal{J}_D$. We first record the kernel identity

$$\bigcap_{T \in \mathcal{J}_D} \ker T = \bigcap_{n=1}^{\infty} \ker T_n.$$

The inclusion from left to right is immediate because $T_n \in \mathcal{J}_D$. Conversely, let $x \in \bigcap_n \ker T_n$. Every finite linear combination of operators of the form AT_n , $A \in \mathcal{B}(E)$, annihilates x . If $T \in \mathcal{J}_D$, choose a net from this linear span converging to T in the weak-star topology. Pairing with $x \otimes \lambda \in E \widehat{\otimes} E^*$ for arbitrary $\lambda \in E^*$ gives $\lambda(Tx) = 0$. Hence $Tx = 0$, proving the reverse inclusion.

Finally,

$$\bigcap_{T \in \mathcal{J}_D} \ker T = D.$$

Indeed D is contained in every kernel by definition of $\mathcal{J}_D$. If $x \notin D$, choose $\lambda \in E^*$ with $\lambda|_D = 0$ and $\lambda(x) = 1$. The rank-one operator $y \mapsto \lambda(y)x$ belongs to $\mathcal{J}_D$, but it does not annihilate x .

Thus $D = \bigcap_n \ker T_n$. Lemma 1 implies that either D or E/D is separable, contradicting the choice of D . Therefore $\mathcal{J}_D$ is not weak-star topologically countably generated. $\square$

Verification and novelty check

The proof is a countable version of the finite argument in Laustsen–White, with the additional observation that countably many separable ranges still have separable closed span. The theorem is strictly stronger than the source’s failure of weak-star topological finite generation for the same ideal.

Bounded novelty checks used the run indexes for ‘1811.02399’, kernel-intersection terms, weak-star Noetherian terms, and countable-generation phrases. Web searches for exact phrases such as “weak-star topologically countably generated”, “weak*-topologically countably generated”, and Laustsen–White/Wark countable-generation variants did not locate this strengthening. A later paper, Arnott–Laustsen arXiv:2503.12940, studies kernels on classical transfinite sequence spaces, but its abstract concerns $\ell_p(\Gamma)$, $c_0(\Gamma)$, and $\ell_1(\Gamma)$, not the Wark-dual weak-star ideal considered here.

Human review focus

The main points to verify are the annihilator identity for countable intersections,

$$\left(\bigcap_n \ker S_n^*\right)_\perp = \overline{\text{span}} \bigcup_n \text{im } S_n,$$

and the weak-star closure argument showing that countable generators force

$$\bigcap_{T \in \mathcal{J}_D} \ker T = \bigcap_n \ker T_n.$$

Both are standard, but they are the exact places where the finite proof is extended.

References

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